

Implementation Note: Critical Design Decisions

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This note highlights four non-trivial design decisions made under practical constraints, focusing on robustness rather than idealized assumptions.

1. Soft Logit Biasing over Constrained Decoding

Instead of enforcing hard vocabulary constraints during ASR decoding, a soft N-gram bias was applied:

$$\tilde{z}_t = z_t + \lambda \log P(w_t \mid w_{t-1}) \quad (1)$$

This preserves acoustic likelihood while guiding decoding toward domain-consistent phrases. Hard constraints were avoided as they often override acoustic evidence, causing systematic errors under noisy or accented speech.

2. Deterministic Translation instead of Neural MT

A rule-based dictionary was used in place of neural translation:

$$w_{\text{bho}} = \begin{cases} \mathcal{D}(w), & w \in \mathcal{D} \\ w, & \text{otherwise} \end{cases} \quad (2)$$

This decision reflects the absence of reliable Bhojpuri MT systems and the need to preserve technical terminology. Neural models were rejected due to instability and domain drift on low-resource language pairs.

3. Prosody Transfer as a Proxy for Speaker Identity

Direct speaker-conditioned synthesis was infeasible with the available TTS model. Instead, speaker characteristics were approximated via prosody alignment using DTW:

$$DTW(X, Y) = \min \sum d(x_i, y_j) \quad (3)$$

Pitch and energy contours were transferred to impose temporal structure. This leverages the insight that perceived speaker identity is strongly influenced by rhythm and intonation, not just spectral features.

4. Segment-Level MCD instead of Global Comparison

Standard MCD assumes phoneme-aligned signals, which does not hold in cross-lingual synthesis. A sliding-window DTW strategy was used to compute MCD over locally aligned segments.

This avoids artificially inflated distortion due to linguistic mismatch and provides a more meaningful estimate of acoustic similarity.

Conclusion

Each design choice prioritizes stability under real-world constraints: low-resource translation, cross-lingual mismatch, and limited model support. The resulting system favors robustness and interpretability over theoretically optimal but impractical solutions.